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Learning-based autonomous navigation, benchmark environments and simulation framework for endovascular interventions

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ABSTRACT

Endovascular interventions are a life-saving treatment for many diseases, but they suffer from drawbacks such as radiation exposure and the potential scarcity of proficient physicians. Robotic assistance during these interventions could be a promising support for these problems. Research focusing on autonomous endovascular interventions using artificial intelligence-based methodologies is gaining popularity. However, variability in assessment environments hinders the comparability of different approaches, primarily due to each study employing a unique evaluation framework. In this study, we present autonomous endovascular instrument navigation based on deep reinforcement learning for three distinct digital benchmark interventions: BasicWireNav, ArchVariety, and DualDeviceNav. The benchmarks focus on aortic arch to supra-aortic navigation, representing fundamental large-vessel navigation skills. The benchmark interventions were implemented with our modular simulation framework stEVE (simulated EndoVascular Environment). Autonomous controllers were trained solely in simulation and evaluated in simulation and on physical test benches with camera and fluoroscopy feedback. Autonomous control for BasicWireNav and ArchVariety reached success rates up to 98/100 in simulation and was successfully transferred to the physical test benches with a success rate of up to 97/100. The experiments demonstrate the feasibility of stEVE and its potential to transfer simulation-trained controllers to real-world scenarios. However, they also reveal areas that offer opportunities for future research. Furthermore, this work reduces barriers to entry and increases the comparability of research on learning-based assistance systems for endovascular navigation by providing open-source training scripts, benchmarks, and the stEVE framework.

1. Introduction

Endovascular interventions are considered the standard of care for a wide range of often critical conditions, including myocardial infarction and large vessel occlusive ischaemic strokes. However, they are not without challenges. Physicians must operate alongside the patient and fluoroscopy equipment, which exposes them to radiation and poses a health risk. Although modern equipment is designed to minimize this exposure, further reductions are desirable [1–3]. Frequent wearing of heavy radiation protection equipment, usually made of lead-impregnated fabric, can lead to orthopedic problems [4]. Furthermore, the complexity of endovascular treatments requires extensive specialized training and equipment. This leads to a potential scarcity of proficient physicians, particularly in remote and underserved regions [5].

Robot-assisted endovascular interventions offer a promising solution to address these challenges faced by physicians [6,7]. Through remote manipulation capabilities, these robotic systems allow operators to remain in radiation-shielded areas while achieving optimal patient

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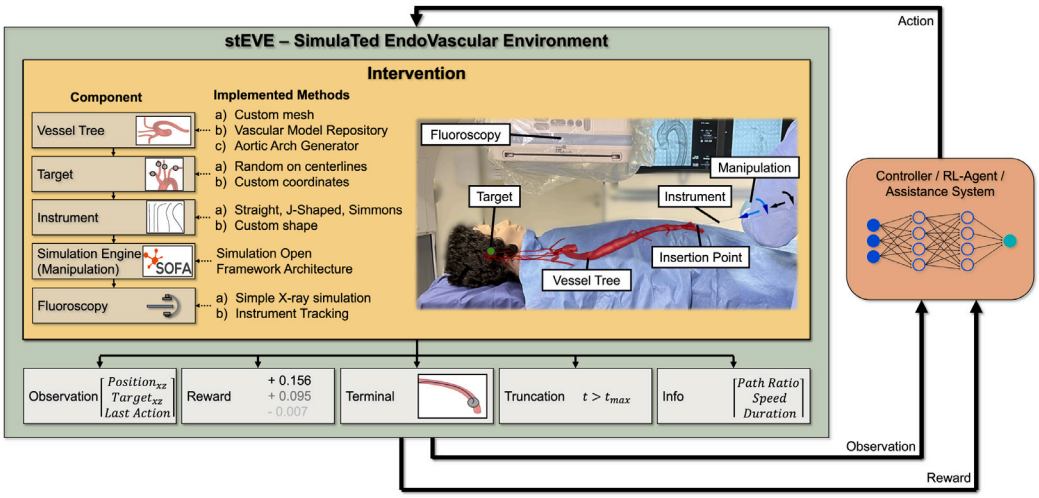


Fig. 1. Structure of stEVE and its components derived from the illustrated endovascular intervention, the reinforcement learning framework and the closed-loop control with a potential assistance system.

positioning that minimizes radiation exposure, unrestricted by the ergonomic limitations of a manual intervention [8]. In addition, they facilitate long-distance remote interventions through telesurgery [9].

The next step in the evolution of endovascular robots is to integrate assistance systems and higher levels of automation to support surgeons during procedures [10]. The potential benefits of automation in interventions include improved patient safety and quality of care, as well as reduced hospital expenditures [11]. Autonomous navigation has significant potential to improve patient care by allowing physicians to focus on critical aspects of intervention planning and supervision, rather than the complex task of instrument manipulation, thus reducing their workload [12]. Moreover, advancements towards autonomous navigation can serve as a foundation for simpler assistance systems, such as training platforms for novice surgeons or assistive guiding systems to enable the surgeon to perform optimal navigation, either digitally or through haptic feedback, akin to lane-keeping assistance in automobiles.

Research on autonomous navigation in endovascular interventions has gained momentum in recent years. Early attempts with conventional control had only limited success [13,14]. Subsequently, the field has shifted towards learning-based approaches, recognizing that endovascular interventions are particularly suitable for the application of machine learning techniques [15].

However, a direct comparison of the effectiveness of these different control algorithms is hampered by significant differences in their evaluation environments. Simulation-based approaches have utilized 2D physics [16] or 3D finite element simulations of realistic [17–20] and synthetic [21,22] vessel systems for evaluation.

In real-world physical applications, some studies [23–26] use phantoms with technically motivated geometries constructed from transparent hard plastic where feedback is received through an overhead camera. Others employ flexible phantoms that replicate the geometries of real patients' vessels, with feedback derived from camera images [27–29] or an electromagnetic tracking system [30]. In another study [31], evaluation was conducted in ex-vivo animal tissue, using fluoroscopy as feedback.

Each study is conducted in its own unique evaluation environment, featuring vessel models with specific geometries and material characteristics. Furthermore, the absence of open-source availability for both training scripts and evaluation setups poses a significant obstacle to reproducibility [32].

Common benchmark interventions are imperative to improve comparability in autonomous navigation in robotic endovascular interventions. If available open-source, these benchmarks can foster research similar to open-source datasets for vision-based assistance systems [33] or the Farama gymnasium environments for reinforcement learning algorithms [34]. Commercial simulators such as the VIST® system (Mentice AB, Sweden) or CATHIS® system (CATHI GmbH, Germany) could serve as benchmarks, but while these systems offer cutting-edge simulation capabilities, they are proprietary solutions that lack the depth of access and availability required for rigorous research purposes.

To provide an initial common simulation-based environment for development and evaluation and enhance comparability across studies, we propose three distinct endovascular benchmark interventions: *BasicWireNav*, *ArchVariety*, and *DualDeviceNav*. These initial benchmarks focus on fundamental navigation challenges currently faced by researchers and are designed to be systematically expanded in the future to address additional complexities. The benchmarks are implemented using our modular framework *stEVE* (***s**imula**t**ed **E**ndo**V**ascular **E**nvironment*), a simulation framework tailored for endovascular interventions that enables flexible adaptation and extension of benchmark scenarios. This modular approach ensures that as algorithmic capabilities advance, the benchmark suite can grow to encompass more sophisticated intervention challenges while maintaining consistency and comparability across studies.

Furthermore, an experimental study is presented that evaluates the real-world applicability of the simulated benchmarks and provides baseline results for autonomous instrument control based on our methodology of [19], complemented by an ablation study that systematically analyzes reward function design to guide future algorithm development.

The key contributions are summarized as follows:

(i) Demonstration of autonomous control capabilities using established reinforcement learning methods that validate successful simulation-to-reality transitions, robust patient-individual generalization, and feasibility assessment of dual-instrument navigation scenarios.[1]
(ii) Novel benchmark interventions for autonomous endovascular navigation to improve comparability and lower the barrier to entry for future studies.[2]
(iii) Development of the modular, open-source framework *stEVE* to advance research in assistance systems for robotic endovascular interventions.[3]

[1] See https://github.com/lkarstensen/stEVE_training for open-source code.
[2] See https://github.com/lkarstensen/stEVE_bench for open-source code.
[3] See https://github.com/lkarstensen/stEVE for open-source code.

Table 1
Comparison of simulation frameworks in autonomous endovascular navigation research.

Research	Engine	Instruments	Vessel system	Open source	Flexibility
Molinero et al. [35]	Unity	catheter	single (aorta)	–	–
Meng et al. [17]	Unity	guidewire	single (aorta)	–	–
Schegg et al. [18]	SOFA	guidewire	various (coronary)	–	–
Kienzlen et al. [21]	SOFA	guidewire	single (synthetic)	–	–
Scarponi et al. [22]	SOFA	guidewire	various (synthetic)	–	–
Jianu et al. [20]	MuJoCo	catheter	single (aorta)	✓	low
stEVE (ours)	SOFA	guidewire & catheters	various (aorta & cerebral)	✓	high

(iv) Systematic ablation study of reward function components that provides insights into reward design for endovascular navigation tasks.

We have chosen a modular approach to ensure flexible use. Our training scripts[1] facilitate the reproduction of our results and provide a low entry barrier for other researchers to set up their own training. By separating the benchmark environments[2] from the training, researchers can implement their own training routines using the reinforcement learning framework of their choice. Additionally, the *stEVE*[3] simulation framework can be utilized for other research in endovascular interventions beyond the scope of autonomous navigation.

2. stEVE — simulated EndoVascular Environment

The software framework stEVE offers a modular toolkit that enables users to flexibly compose simulated interventions according to their clinical needs and technical requirements. Furthermore, the simulated intervention data can be seamlessly converted to feedback suitable for reinforcement learning. The components of stEVE, their existing implementations, and the embedding with an assistance system are shown in Fig. 1.

An overview of simulation frameworks for research on autonomous endovascular navigation is given in Table 1. In comparison, stEVE offers a wider range of instruments and vessel systems, is open source available, and exhibits a higher degree of modularity and extendability. Especially the last point is an advantage in the adoption of this framework for future research.

The practical utility of stEVE has been validated through the deployment across three independent research institutions: Fraunhofer IPA (Mannheim, Germany), FAU SPARC Lab (Erlangen, Germany), and KCL BMEIS (London, UK). Each group has successfully adapted the modular components to their research requirements while maintaining computational reproducibility between institutions.

In the subsequent sections, we explore the medical background, derive the components of a simulated intervention, and explore the reinforcement learning interface.

2.1. Medical background

During endovascular procedures, instruments are typically introduced into the vascular system at specific sites, including the femoral artery (located in the groin), the brachial artery (at the elbow), or the radial artery (at the wrist). Subsequently, these instruments must be skillfully navigated through the vascular network by probing the correct vessel at bifurcations to the target region for further treatment, e.g. mechanical thrombectomy in cerebral vessels in the context of stroke. This navigation process is achieved by external manipulation of the proximal end of the instrument to facilitate the entry of the instrument tip, the distal end, into the target artery [36].

Endovascular navigation can be systematically divided into two distinct phases: initial large vessel navigation and small vessel navigation in the target artery system. Large-vessel navigation encompasses instrument manipulation through major arteries to reach the target artery system, such as traversing the aortic arch to access supra-aortic vessels. The primary challenge is precise directional control at vessel bifurcations to enter the correct branch vessel [37]. Small-vessel navigation requires advancing instruments through progressively smaller vessels to reach the treatment site. The critical challenge is to achieve effective control of the distal tip, specifically the transmission of torque and force, through tortuous anatomy [38].

In many cases, simultaneous manipulation of two concentric instruments (e.g., guidewire and catheter) becomes necessary, as they mutually influence the combined stiffness and shape of the instrument tip utilized for navigation. Although sometimes several instruments are inserted in parallel at the same insertion point, e.g. for fenestrated endovascular aortic aneurysm repair, or at different insertion points, e.g. for treatment of aortic bifurcation stenoses, most interventions utilize a concentric arrangement of instruments at one insertion point [39].

Throughout this intricate navigation procedure, continuous feedback is provided through fluoroscopy imaging, a low-dose X-ray modality that provides real-time visualization of the instrument positions inside the vasculature. When the instruments reach the target point, navigation is successful, and treatment of the lesion can begin [40]. In this work, we consider the navigation aspect of the intervention.

The modular design of stEVE directly addresses the common challenges faced by researchers working with digital assistance systems in endovascular interventions. The key features that define an endovascular intervention and that consequently can be adjusted in stEVE for various research applications are: (1) Patient-specific anatomy: The individual patient and the disease define both the location of the target site and the geometry of the vessel to be navigated. (2) Procedural decisions: The physician's choice of insertion point and instrument selection based on patient anatomy and target accessibility. (3) Imaging infrastructure: The intervention room's imaging capabilities, such as single- or bi-plane fluoroscopy systems with varying angulations.

2.2. Components of the intervention within stEVE

stEVE comprises five modular components to define an intervention designed to accommodate diverse research requirements: (1) Vessel Tree Module — allowing individual vessel geometry and insertion point definition; (2) Target Definition Module — enabling target region definition; (3) Instrument Module — providing standard or individual instruments; (4) Imaging Module — offers configurable fluoroscopy systems; and (5) Simulation Engine — utilizing the SOFA framework ensuring compatibility and extensibility for future developments.

2.2.1. Vessel tree with insertion point

Considering the unique geometry of the vessel system of each patient, it is essential that the vessel tree module is designed for easy adaptation. In addition, physicians determine the optimal access site based on the planned intervention, the anatomy of the patient, and the respective target area. Consequently, it is essential that the insertion point can be defined flexibly for the intervention.

The vessel tree defines the geometric characteristics of the vascular system in the form of an ordered centerline point cloud and a surface mesh representing the vessel walls. In addition, an insertion point is specified that consists of both the position and the direction. Three

different methods are currently provided for the integration of vessel trees: Loading a custom vessel mesh from any anatomy, utilizing vessel models offered by the vascular model repository [41], or generating parameterized artificial aortic arches of six different types of aortic arches [19].

2.2.2. Target site

For each procedure, a navigation target must be defined and possibly modified during the intervention. Depending on the specific intervention and the patient's vessel geometry, targets can either be manually selected by the user or automatically chosen in the target vessel. The target is considered reached when the tip of any instrument approaches a predefined threshold distance from the target location.

2.2.3. Instruments

The instruments used in interventions exhibit various mechanical characteristics and shapes. The choice of instrument depends on the geometry of the vessel, the target within that geometry, and the physician's preference. Although the majority of instruments are readily available off-the-shelf, there are instances where the physician customizes the instrument, such as coronary guidewires. Hence, the capability to define arbitrarily shaped instruments within the simulation framework is necessary.

stEVE offers comprehensive instrument modeling capabilities through two complementary approaches: *geometric flexibility* and *physical property customization*. For geometric design, the framework allows the creation of instruments of any shape by connecting straight segments and segments curved in one (i.e., bend) or two (i.e., helix) dimensions. Based on this approach, stEVE has implemented parameterized standard instruments with a straight or angled tip, as well as the Simmons catheter [42] with customizable dimensions and physical properties.

For physical property modeling, each instrument can be defined with up to two distinct regions having different mechanical properties, typically representing a stiff proximal shaft and a soft distal tip. This dual-region approach reflects clinical reality, where instruments feature stiffer proximal sections for pushability and torque transmission, combined with flexible distal tips for safe navigation and reduced vessel trauma. Each region can be independently configured with the physical parameters inner diameter, outer diameter, Poisson ratio, Young's modulus, and mass density.

2.2.4. Fluoroscopy

An adaptable fluoroscopy feedback mechanism is essential, supporting configurations such as single-plane or biplane imaging systems [43]. stEVE provides the capability to instantiate simulated fluoroscopy systems as needed. Depending on the context, multiple systems with different viewing angles can be applied. Currently, two types of fluoroscopy feedback are implemented: a simulated tracking algorithm with instrument coordinates as output and a basic fluoroscopy simulator, which takes the instruments' positions and creates a fluoroscopy-like image by inserting the instruments in a noisy grayscale image.

2.2.5. Simulation engine

To simulate an intervention, that is movement of the instrument within the specified vessel tree, stEVE relies on the SOFA (Simulation Open Framework Architecture) simulation engine [44] with the BeamAdapter plugin [45]. SOFA represents the prevailing standard in the endovascular simulation research community, as evidenced by its adoption in the majority of current studies seen in Table 1, and provides open-source accessibility essential for reproducible research.

The choice of SOFA over alternative physics engines offers specific technical advantages for endovascular simulation. While simulators like CathSim utilize MuJoCo, which excels in rigid body and contact simulation, SOFA's focus on soft-body physics provides advantages for modeling flexible endovascular instruments (guidewires, catheters) that exhibit significant deformation during navigation. The finite-element method (FEM) implementation models vessel walls as meshes and instruments as chains of 1-dimensional elements based on Kirchhoff's rod theory, enabling realistic simulation of instrument bending, torsion, and vessel-instrument interactions.

2.3. Reinforcement learning interface

The prevailing standard interface for reinforcement learning is set by the Farama Gymnasium [34], which describes a structured interface as

observations,	e.g., instrument and target position,
rewards,	e.g., $0.001 \cdot \Delta$(distance to target),
terminal states,	e.g., target reached,
truncations,	e.g., timeout, and
infos,	e.g., current translation speed.

stEVE offers a suite of tools designed to facilitate the conversion of intervention data to the Gymnasium interface. The interface definition can be based on the full intervention data, that is, vessel geometry, three-dimensional target position, instrument shape and characteristics, three-dimensional position data of each instrument, and fluoroscopy images.

It is imperative for the user to keep in mind the accessibility of the real-world data, as access to the full intervention data in the simulation differs from the constraints encountered in a real-world intervention. For example, the simulation provides three-dimensional position information for each instrument, while from a single fluoroscopy image only two-dimensional positions are available in the real world. However, the reward function, only required during training, can leverage the data-rich environment of the simulation in cases where the design and training is performed in the simulation even if the evaluation is performed in the real world. However, observations, required during evaluation, must be carefully tailored and adjusted to account for the real-world limitations encountered during the evaluation and application phase.

3. Benchmark environments

Based on the medical background, we propose three distinct benchmark environments, utilizing the stEVE framework. Each benchmark focuses on distinct facets of instrument navigation during endovascular interventions: fundamental maneuvering of the guidewire, adaptability to patient-specific variations, and concurrent handling of two concentric instruments, the guidewire and the catheter.

The proposed benchmarks are designed to systematically evaluate the fundamental navigation capabilities identified in the medical background. We focus on large-vessel navigation from the aortic arch to the supra-aortic arteries, which represents the prevailing standard evaluation scenario in the autonomous endovascular navigation research community [32]. This anatomical region provides a well-characterized testbed with consistent bifurcation patterns and moderate tortuosity, enabling reproducible evaluation across different algorithms.

While our current implementation focuses on neurovascular access routes, the fundamental navigation principles (bifurcation selection, tip control, multi-instrument coordination) are directly applicable across endovascular specialties including cardiovascular, peripheral vascular, and interventional radiology procedures.

In all three benchmarks, the instruments are angled with a tip radius of 12.1 mm and a tip angle of 0.4π radians, and the velocities are limited to 35 $\mathrm{mm/s}$ translational and 3.14 $\mathrm{rad/s}$ rotational speed. Tracking data is utilized as feedback, rather than fluoroscopy imaging. Most studies use tracking data as input for their control algorithm, either from an electromagnetic tracking system or a tracking algorithm that analyzes images [32]. Simulating the tracking algorithm instead of images ensures that the performance of the tracking system does not

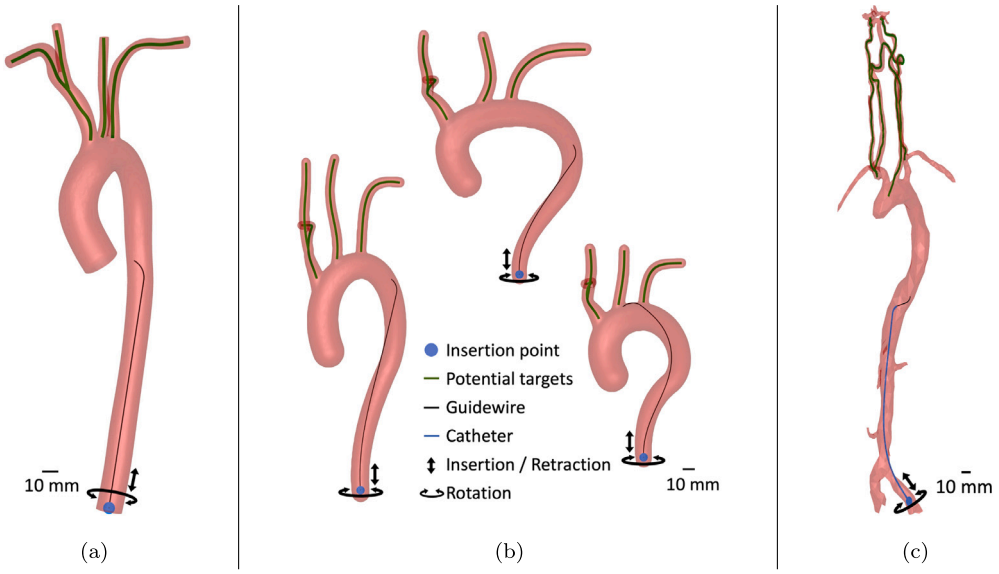


Fig. 2. Proposed benchmark interventions: (a) *BasicWireNav*: single real aortic arch, single instrument and supra-aortic targets. (b) *ArchVariety*: multiple artificial aortic arches, single instrument and supra-aortic targets. (c) *DualDeviceNav*: single real vascular system, dual instruments and targets in the common carotid and vertebral arteries.

impact the performance of the control algorithm. For this purpose, the simulated tracking algorithm extracts the instrument position from the simulation and transforms them into a coordinate system of an right anterior oblique projection image plane. The imaging frequency is set at 7.5 Hz, a typical value for optimized radiation reduction [46]. Detailed physical properties and parameters for all benchmark configurations are provided in Appendix A.

3.1. BasicWireNav

This benchmark evaluates fundamental navigation capabilities through single-instrument control in a fixed anatomical environment. It assesses the core skill of directional control at vessel bifurcations — the primary challenge in large-vessel navigation. The vascular model is derived from a healthy 23-year-old female from the vascular model repository (#0011_H_AO_H) [41]. The navigation target is randomly selected for each episode along the centerline of the brachiocephalic trunk, carotid, or subclavian arteries, resulting in 562 different potential targets. Fig. 2(a) depicts the navigation task.

3.2. ArchVariety

This benchmark evaluates generalization capabilities across varying anatomical configurations, testing algorithmic robustness to patient-specific variations in vessel geometry while maintaining the same fundamental navigation challenge as *BasicWireNav*. Vascular models consist of randomly generated type I aortic arches. The navigation target is randomly selected for each episode along the centerline of the brachiocephalic trunk, carotid, or subclavian arteries. Fig. 2(b) provides an illustration of the navigation task for three exemplary aortic arches.

3.3. DualDeviceNav

This benchmark evaluates advanced navigation capabilities by incorporating both large-vessel and small-vessel navigation phases. The concurrent manipulation of guidewire and catheter tests coordinated multi-instrument control essential for complex endovascular procedures. The patient's vascular model is manually segmented from a computed tomography angiographic scan of the aorta and a magnetic resonance angiography scan of the cerebral vessels. Subsequently, these segments were manually assembled into a cohesive model. Navigation targets are randomly placed along the centerline of the common carotid or vertebral arteries, resulting in 898 potential targets. Fig. 2(c) shows an illustration of the vessel system and the navigation task.

This benchmark represents an intentionally challenging scenario designed to explore the limits of current reinforcement learning methodologies and identify areas requiring future research development. The concurrent manipulation of guidewire and catheter involves complex physical interactions including nested instrument dynamics, and coordinated tip control.

4. Autonomous control

To validate the feasibility of the benchmarks and demonstrate simulation-to-reality transferability, an autonomous navigation study is performed. Additionally, an ablation study analyzes the effect of individual reward components on training success, providing insights into reward design for endovascular navigation tasks.

The autonomous navigation study serves two critical purposes for establishing these benchmarks as a common evaluation framework. First, it provides baseline results demonstrating the feasibility and proficiency of learning-based approaches on each benchmark task. Second, and equally important for benchmark validation, it evaluates the realism of the simulation environment through simulation-to-reality transfer experiments. Successful sim-to-real transfer indicates that the simulation accurately captures the essential physics and challenges of real endovascular navigation. Large discrepancies between simulation and real-world performance would indicate fundamental simulation limitations that could deter adoption by the research community. Notably, our simulation does not incorporate artificial inaccuracies or domain randomization techniques, like stochastic tracking errors, that might mask simulation-to-reality gaps, providing a direct assessment of the benchmark's representativeness of real-world navigation challenges.

For each benchmark, a separate autonomous reinforcement learning-based controller is designed and trained in the simulation environment. BasicWireNav and ArchVariety controllers are additionally evaluated on physical test benches featuring camera and fluoroscopy feedback to establish simulation-to-real transferability. For the DualDeviceNav benchmark, we deliberately apply the same reinforcement learning methodology used for the simpler navigation tasks, despite the increased complexity of multi-instrument control. This methodological consistency serves a specific research purpose: to systematically evaluate the limits of current state-of-the-art approach when faced with

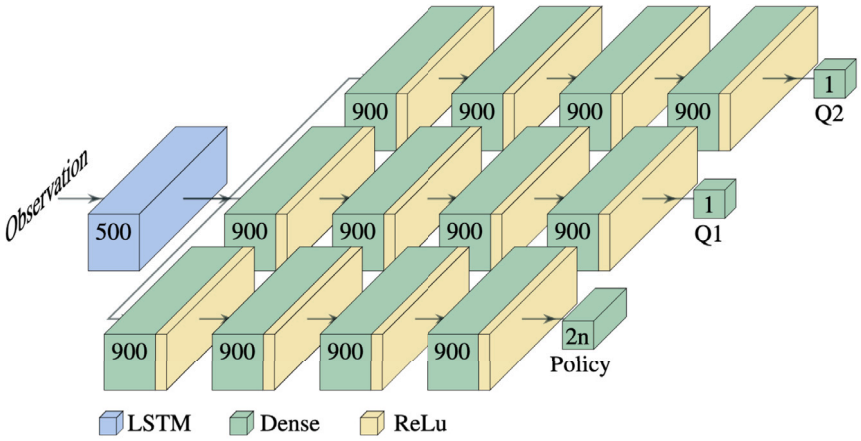


Fig. 3. The structure of the neural networks of the soft-actor-critic agent with input embedder, Q1- and Q2-network with one output node and policy-network with $2n$ output nodes, where n is the amount of instruments. Amount of layers and nodes is exemplary for the *BasicWireNav* controller.

more complex coordination challenges. By intentionally applying the methodology beyond its optimal domain, we aim to establish baseline performance limits and identify specific failure modes that guide future algorithmic development.

The ablation study systematically evaluates reward function design by training seven different controllers on BasicWireNav, each with different combinations of reward components. This analysis provides quantitative insights into the contribution of each reward term and offers guidance for reward design in endovascular navigation tasks, addressing a key challenge in reinforcement learning applications where reward engineering significantly impacts training success.

In this section, we detail our controller architecture and approach to training, as well as the evaluation modalities.

4.1. Controller architecture and training scheme

The controller applies the soft-actor-critic (SAC) method [47] and adapts the neural network architecture, reinforcement learning methodology, and observation and reward design from our previous work [19]. While the fundamental algorithmic approach remains consistent with our prior research, this work contributes novel benchmark environments, adapted training parameters for the new scenarios, extended evaluation including real-world validation, and systematic reward design analysis. The focus of this study is not algorithmic novelty but rather the establishment of common benchmark environments and demonstration of their validity through comprehensive simulation-to-reality evaluation.

The structure of the neural networks is illustrated in Fig. 3. The observation is processed by an observation embedder of long-short-term memory (LSTM) nodes without an activation function. The embedded observation is subsequently fed to the policy-, Q1- and Q2-network. Each of them consists of strictly feed-forward densely connected hidden layers with a rectified linear unit (ReLu) as activation function. For the Q1- and Q2-network, the output layer has 1 neuron representing the Q-value, which is solely utilized for training of the neural network. The output layer of the policy-network has $2 \times n_{instruments}$ neurons representing the mean value and standard deviation of the translation and rotation speed for each instrument. The illustrated number of layers and nodes for the observation embedder and hidden layers are exemplary for the *BasicWireNav* controller.

The observation provided to the controller consists of four components: the current and previous position of the instrument tip, the target position, and the last action executed. The tip position is defined by three points equally spaced 2 mm apart along the instrument starting from the tip. Each point of the instrument tip, as well as the target position, is given as two-dimensional coordinates in the image plane of the imaging system. The action is provided as translational and rotational speed at the insertion point for each instrument. Each observation is normalized to the range $[-1, 1]$ where -1 corresponds to the minimum value and 1 to the maximum value of the observation element in the navigation task.

The reward (R) per step is defined as follows:

$$R = R_{step} + R_{path} + R_{target}$$

with

$$R_{step} = -0.005$$

$$R_{path} = 0.001 \cdot \Delta pathlength$$

$$R_{target} = \begin{cases} 1.0 & \text{if target reached.} \\ 0 & \text{otherwise.} \end{cases}$$

where *pathlength* represents the spatial distance between the instrument tip and the target along the vessel centerlines in millimeters.

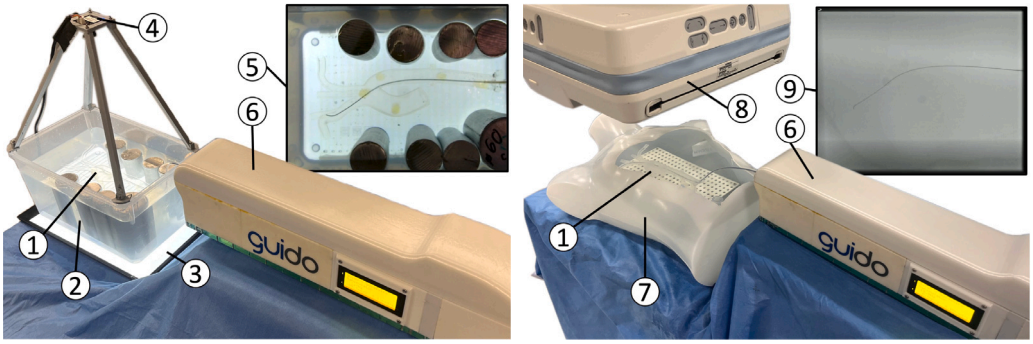
An episode corresponds to one navigation task from the insertion point to the target and is considered successful if the target is reached within a 5 mm threshold. The episode is deemed unsuccessful if a time limit is reached, the instrument tip reaches the end of an incorrect vessel branch, or a simulation error occurs. The control frequency matches the imaging frequency.

Training is performed solely in simulation and is carried out for 1.0×10^7 exploration steps, that is, cycles of the control loop, as preliminary experiments have shown that this is sufficient for the controller to reach its maximal success rate. To monitor training progress, the controller performs 100 evaluation episodes every 2.5×10^5 exploration steps.

4.2. Evaluation

We employ three distinct evaluation modalities: (i) in simulation, (ii) a test bench with camera-based feedback (TC), and (iii) a test bench with fluoroscopic feedback (TF).

TC, as depicted in Fig. 4(a), incorporates a 3D-printed transparent phantom (Clear, Formlabs Inc., USA) submerged in glycerin. Glycerin is utilized to improve visibility of the instrument inside the phantom because of the similarity of its refractive index to that of the print resin. An overhead camera (SUSB1080P01-LC1100, Ailipu Technology Co., China) and a lower-mounted light source provide feedback. A tracking algorithm is utilized to extract instrument positions from the camera image [48]. A custom guidewire manipulator executes movements



(a) (b)

Fig. 4. (a) Test bench with camera feedback in phantom ① in a specimen container filled with glycerin ② on a light source ③, a camera mounted above ④ which generates real images ⑤. Images are processed by a tracking algorithm and the controller calculates new actions based on instrument tracking (not depicted). Actions are performed by a guidewire manipulator ⑥. (b) Test bench with fluoroscopy system ⑧ generating real images ⑨ and water-filled mannequin as specimen container ⑦.

(i.e. translation and rotation) to the guidewire (Radifocus Guide Wire M Stiff Type, Angled, 0.035", Terumo Corporation, Japan) instructed by the controller. The manipulator has a stepper motor to which the guidewire is attached directly for infinite rotation. The guidewire and stepper motor are mounted on a linear rail driven by a second stepper motor, allowing 300 mm of translational movement.

TF, as shown in Fig. 4(b), mirrors TC, with the phantom, tracking algorithm, manipulator, and guidewire being identical. However, in this case, the phantom is placed within a water-filled mannequin, which mimics uniform surrounding tissue and a medical fluoroscopic system (Artis Zeego, Siemens Healthcare GmbH, Germany) provides image feedback during the intervention.

TC facilitates extensive experimentation on the test bench because of the accessibility and user-friendly nature of its components. TF adds an extra layer of realism by incorporating fluoroscopic feedback. The variance in imaging modalities introduces discrepancies in tracking accuracy, primarily attributable to distortion effects, particularly noticeable for movements perpendicular to the image plane.

The test bench environments represent static conditions without pulsatile flow, designed specifically for validation of the simulation framework rather than comprehensive clinical replication. While pulsatile blood flow would enhance clinical realism, the current static setup enables controlled evaluation of the navigation challenges captured in the simulation benchmarks. The test bench configurations prioritize reproducible assessment of simulation-to-reality transfer for the core navigation skills rather than full physiological complexity.

The simulation-to-reality evaluation serves as benchmark validation rather than merely algorithm demonstration. For these benchmarks to serve as a common evaluation environment, the simulation must demonstrate sufficient realism to ensure that algorithmic performance translates meaningfully to real-world scenarios. Poor sim-to-real transfer would indicate fundamental limitations in the simulation framework that could mislead algorithm development and comparison.

For *BasicWireNav* training episodes are capped at 20 s. This duration was determined to be sufficient for navigation tasks within the velocity constraints, based on manual pre-trials. The evaluation episodes extend to 120 s, since the method of short training and long evaluation episodes facilitates quicker learning during training and allows ample time to rectify navigational errors during evaluation. Training is performed with an observation embedder of 500 neurons, four layers of 900 densely connected neurons for Q1-, Q2-, and policy-network and a learning rate of 2.199×10^{-4}. Evaluations are carried out in simulation and on the TC for 100 targets, with the initial 10 out of 100 targets being additionally assessed on the TF. Evaluation on the TC examines simulation to reality transfer and the limited scope of evaluation on TF, focusing on just 10 targets due to availability constraints of the fluoroscopic equipment, is designed to examine the transition from camera-based to fluoroscopy-based feedback.

For *ArchVariety*, episodes follow the same time limits as *BasicWireNav*. Training is performed with an observation embedder of 900 neurons, three layers of 400 densely connected neurons for Q1-, Q2-, and policy-network and a learning rate of 3.218×10^{-4}. For evaluation, the aortic arch with "661023725" as seed for the random number generator is sampled, and 100 targets are navigated both in simulation and on TC. This evaluation setup is designed to assess simulation-to-reality transfer for a different scenario.

For *DualDeviceNav*, training episodes are limited to 66 s, while the evaluation episodes last 133 s. Training is performed with an observation embedder of 500 neurons, four layers of 900 densely connected neurons for Q1-, Q2-, and policy-network and a learning rate of 2.199×10^{-4}. The evaluation is carried out exclusively in simulation for 100 targets, offering a controlled and stable environment ideal for benchmarking purposes.

In the reward ablation study seven different controllers are trained using the benchmark BasicWireNav with the same training procedure and hyperparameters as the main autonomous control experiment. The seven reward configurations are: (i) Complete reward (baseline): $R = R_{step} + R_{path} + R_{target}$, (ii) No target reward: $R = R_{step} + R_{path}$, (iii) No step penalty: $R = R_{path} + R_{target}$, (iv) No path progress: $R = R_{step} + R_{target}$, (v) Path progress only: $R = R_{path}$, (vi) Success reward only: $R = R_{target}$, and (vii) Step penalty only: $R = R_{step}$. Training performance is evaluated every 2.5×10^5 exploration steps using 100 episodes with a fixed set of randomly selected targets. Training continues until the controller achieves 100% success rate or until convergence plateaus. To ensure consistent comparison across all reward variants, the same set of 100 target locations is used for evaluation across all seven training runs, though these targets differ from those used in the simulation-to-reality transfer study.

5. Results

Table 2 presents the results of the evaluation experiments, with the key metrics *Success Rate*, which quantifies the number of targets achieved out of the attempted ones; *Path Ratio*, a measure of the remaining distance to the target in unsuccessful episodes, calculated by dividing the remaining distance by the initial distance; and *Duration* of the successful episodes, measuring the time from the start of navigation until the target is reached.

In the *BasicWireNav* experiments, result metrics are given for 100 targets for the simulation and TC experiments. In addition, results metrics are provided for the first 10 of the 100 targets for simulation, TC, and TF to allow analysis of transfer to fluoroscopy on the physical test benches.

During training the controller achieved its maximum success rate after 46 600 episodes, which was achieved after 6 h39 min with 55

Table 2
Results of the Benchmark experiments for 100 trials in simulation. *BasicWireNav* additionally evaluated for 10 trials on the camera (TC) and fluoroscopic (TF) test benches.

Benchmark	Evaluation	Success rate		Path Ratio [%] *failed*		Duration [s] *successful*	
BasicWireNav	Simulation	98/100	10/10	94.3	–	9.6	9.8
	TC	97/100	9/10	81.6	87.4	13.6	18.1
100 trials *10 trials*	TF	–	10/10	–	–	–	19.1
ArchVariety	Simulation	90/100		73.5		19.3	
	TC	84/100		75.9		22.0	
DualDeviceNav	Simulation	40/100		67.8		45.8	

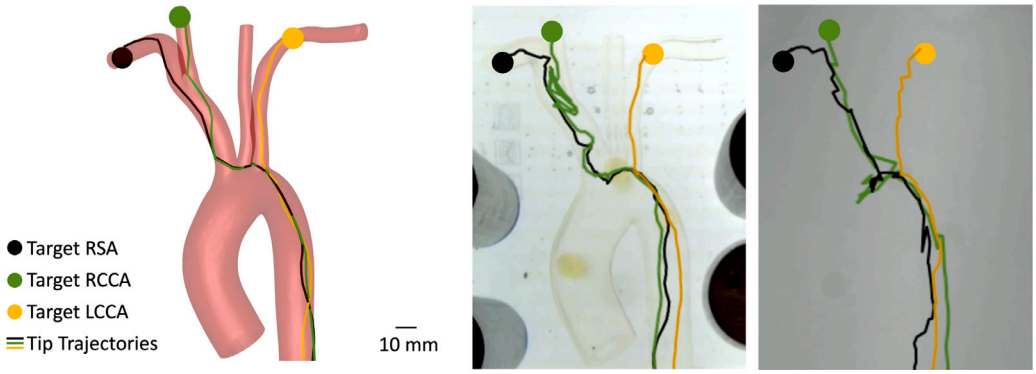


Fig. 5. Trajectories of *BasicWireNav* navigations to targets in the right subclavian artery (RSA) and right and left common carotid artery (RCCA and LCCA) for (a) simulation, (b) TC and (c) TF.

worker agents and one trainer agent on a system equipped with an AMD Threadripper 3970X CPU, NVIDIA RTX 3090 GPU and 128 GB RAM.

For the full 100 targets, the success rates reach 98 out of 100 in the simulation and 97 out of 100 on the TC (p-value 0.653). The path ratio exhibits a significant decline from 94.3% in the simulated environment to 81.6% on TC (p-value 0.045). The duration of successful episodes increases from 9.6 s in simulation to 13.6 s on TC (p-value 0.011).

For the reduced set of 10 targets, the autonomous navigation reaches 10/10, 9/10, and 10/10 in simulation, on TC, and on TF, respectively (p-value 0.583). The path ratio cannot be calculated for the simulation and TF, as no episodes resulted in failure. On TC the failed attempts navigated 87.4% of the path. The duration of the navigation on TC and TF insignificantly increased from 18.1 s to 19.1 s (p-value 0.323).

Trajectories leading to targets in the right subclavian artery (RSA) and the right and left common carotid arteries (RCCA and LCCA) are shown in Fig. 5 for simulation, TC and TF. In simulation, the three targets are successfully reached without complications within a maximum duration of 11 s. On the test benches, the RSA target is reached on the initial attempt, while for the RCCA and LCCA targets, up to 82 s were required to probe the desired artery successfully.

In the *ArchVariety* experiments, the controller reached the maximum success rate after 33700 episodes of training. Training lasted 10h40 min with 29 worker agents and one trainer agent on a system equipped with an AMD Ryzen 5950X CPU, NVIDIA RTX 3070 Ti GPU and 64 GB RAM. In the evaluation, a success rate of 90 out of 100 is achieved in simulation, slightly reduced to 84 out of 100 on the camera-based test bench (p-value 0.209). The path ratio remains similar, at 73.5% and 75.9%, respectively (p-value 0.713), while the duration increases from 19.3 s in simulation to 22.0 s on the test bench (p-value 0.409).

During the *DualDeviceNav* training, the controller reached the maximum success rate after 40800 episodes. Training lasted 71h17 min with 16 worker agents and one trainer agent on a system equipped with an AMD Ryzen 7950X CPU, NVIDIA RTX 4090 GPU and 64 GB RAM. The experiments achieve a success rate of 40 out of 100, with a path ratio of 67.8% for unsuccessful episodes and a duration of 45.8 s for successful episodes.

The reproducibility of these simulation results has been independently verified across multiple research institutions (Fraunhofer IPA, FAU SPARC Lab, KCL BMEIS), with similar computational outcomes achieved when using the same benchmark configurations and training parameters. Minor differences are inherent to the usage of reinforcement learning with stochastic initialization of training weights and random actions chosen for exploration.

During the ablation study, all seven controllers eventually achieved 100% success rate, but exhibited significantly different learning dynamics as shown in Fig. 6. The reward variants can be categorized into two distinct performance groups based on their learning efficiency. Controllers trained with individual reward components — R_{step}, R_{path}, and R_{target} — demonstrated delayed learning onset, with success rates beginning to rise after $3-4\times10^6$ exploration steps and reaching 100% success after $5.75-7\times10^6$ steps. In contrast, controllers incorporating two or more reward components showed accelerated learning, with success rates increasing after $1.5-2\times10^6$ exploration steps and achieving 100% success within $4-5.25\times10^6$ steps. This faster learning group includes the complete reward formulation and all two-component combinations, demonstrating a potential reduction in training time of approximately 25%–40% compared to single-component approaches.

6. Discussion

In this work, we present the modular framework *stEVE* to compose simulated endovascular environments and its ability to represent a diverse range of interventions, introduce three benchmark environments as a crucial first step to make the research and development of autonomous endovascular navigation more comparable and perform experiments in simulation and on real test benches.

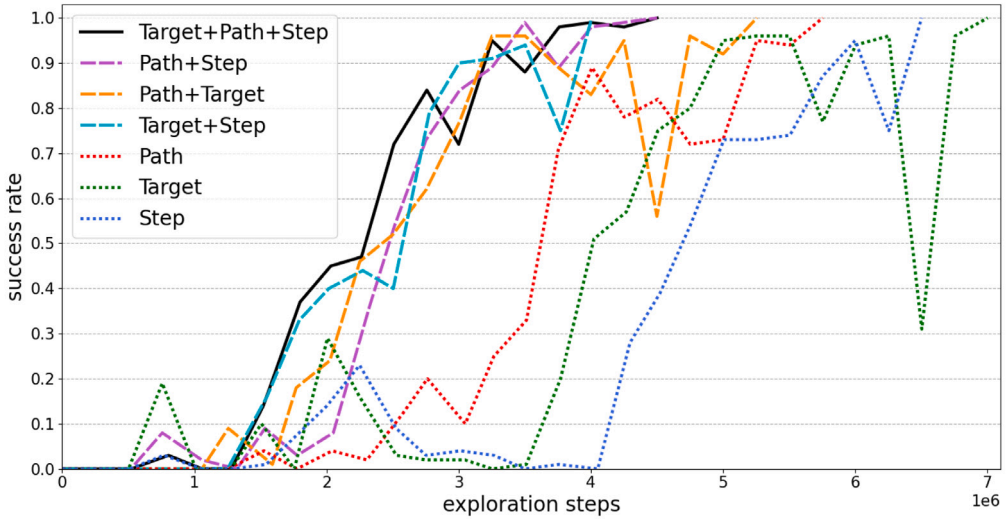


Fig. 6. Success rate of each controller during training for the reward ablation study.

6.1. Autonomous control experiments

The experiments validate both algorithmic capability and, critically for benchmark establishment, the realism of our simulation framework. The successful simulation-to-reality transfer with minimal performance degradation (98/100 to 97/100 success rate for BasicWireNav) demonstrates that our benchmarks capture the essential challenges of real endovascular navigation without requiring artificial domain adaptation techniques. This validation is essential for establishing these benchmarks as a common evaluation platform — significant sim-to-real performance gaps would indicate simulation limitations that could mislead algorithm development and comparison across research groups. However, an increase in episode duration can be observed, presumably attributed to discrepancies between the test bench setup and its simulation. In simulations, the tracking coordinates are accurate, but a real tracking algorithm inherently has noise and inaccuracies. Furthermore, real images, captured via a point source, include distortions when vessel structures span varying image depths. An analysis of the trajectories in Fig. 5 supports this theory, as repeated artery probing attempts are only required on the test benches, while in the simulation arteries are probed on the first attempt. This underlines the robustness of the control algorithm, however, also shows potential to improve realism of the simulation. Integrating noise and depth distortion in the simulation in future work could improve the outcomes. Alternatively, exposing the controller to varying vessel geometries during training enhances robustness in the transfer from simulation to reality, as the increase in duration of the *ArchVariety* experiments is not significant.

The test benches replicate the implemented benchmark conditions, with validation performed in static environments without physiological dynamics such as pulsatile flow or respiratory movement. While pulsatile blood flow would enhance clinical realism, it is technically challenging to implement in simulation due to complex interactions between fluid dynamics and soft body mechanics. For early-stage algorithm development and comparison, simulation-based benchmarks remain essential for establishing common evaluation environments. Future studies should incorporate pulsatile flow and other real-world conditions to advance towards more clinically representative validation. Additionally, clinical deployment will require addressing various domain gap challenges across physics modeling, sensing systems, and environmental factors. Appendix B provides a comprehensive analysis of potential domain gaps and proposed solutions including domain randomization techniques, robust training strategies, and adaptive methods to bridge the simulation-to-reality gap.

The reward ablation study demonstrates that while each individual reward component (R_{step}, R_{path}, and R_{target}) is sufficient for successful training, combining at least two components accelerates learning by 25%–40%. Notably, the complete three-component reward formulation does not provide additional benefits over two-component combinations, and the specific choice of two-component pairing does not substantially affect training efficiency. From a practical implementation perspective, the combination of R_{step} and R_{target} offers significant advantages as both represent sparse reward signals that are computationally efficient to calculate. In contrast, R_{path} requires dense feedback computation at each step, necessitating real-time determination of instrument tip position and shortest path distance to the target along vessel centerlines. This computational overhead becomes particularly problematic when working with recorded clinical intervention data, where calculating dynamic path distances is substantially more complex than simply determining target achievement. Therefore, omitting R_{path} not only maintains training efficiency but also facilitates the practical application of these methods to real clinical datasets, lowering the barrier for clinical translation.

For algorithmic comparisons, Moosa et al. [49] conducted a comprehensive benchmarking of the four reinforcement learning algorithms Deep Deterministic Policy Gradient (DDPG), Twin-Delayed Deep Deterministic Policy Gradient (TD3), Soft Actor-Critic (SAC) and Proximal Policy Optimization (PPO) on the ArchVariety benchmark. In contrast to the recurrent observation embedder of this work, Moosa et al. utilize the standard feedforward versions of the algorithms implemented by Stable Baslines 3 [50]. Their results demonstrate clear performance differentiation across algorithms: success rates ranged from 66%–84% with PPO achieving highest performance after tuning. Notably, these standard implementations achieved lower performance than the custom SAC approach with observation embedder achieving 90% success rate, highlighting the importance of domain-specific algorithmic adaptations.

6.2. Benchmark environments

The benchmark environments are a crucial step in making research and development of autonomous endovascular navigation more comparable. As demonstrated in the experiments, these benchmarks are suitable for learning-based control schemes and serve as basis for further research aimed at improving control algorithms including important skills for the navigation of endovascular instruments.

Benchmark *BasicWireNav* checks the basic manipulation of a guidewire, and our controller showed the ability to solve this task in

simulation and in the real world. Benchmark *ArchVariety* checks patient individual adaptability of the vessel geometry. Our controller demonstrated favorable results with the given training procedure, but still left room for refinement. Future research could include features specifically tailored for generalization, for instance, geometry information from pre-interventional imaging. Benchmark *DualDeviceNav* addresses the challenging problem of concurrent manipulation of two concentric instruments over a long navigation route. The experiment resulted in a moderate but promising success rate of 40 out of 100 and shows the need for further research. Detailed analysis of the 60 failed attempts reveals three primary failure modes: (1) failure to probe the supra aortal vessel, where instruments remain in the aorta (78% of failures), (2) probing of the wrong branch (8% of failures), and (3) failure to advance beyond the descending aorta (14% of failures). Furthermore, in 67% of the cases the catheter remained at the insertion point, not partaking in the navigation attempt and in 12% of the cases the guidewire got entangled by forming loops in the aorta which cannot be observed by the current observation design. Qualitative analysis of navigation attempts revealed that the further-inserted instrument exhibited expected movements, while the shorter instrument displayed chaotic motion. This discrepancy can be attributed as the observation focuses solely on the tip of the more deeply inserted instrument, neglecting information about the shorter one. The observation limitation is particularly problematic at vessel bifurcations, where coordinated instrument positioning is critical for successful navigation but cannot be adequately captured by monitoring only the lead instrument tip.

To enhance performance, potential methods include integrating additional information about individual instruments into the observation, such as the insertion length and relative positioning between instruments, or refining the reward system to penalize unfavorable relative positions and reward coordinated movements. Additionally, enhancing the underlying reinforcement learning method, such as adopting a multi-agent approach where separate policies control each instrument while maintaining coordination constraints, could prove beneficial for this multi-instrument control problem.

The presented benchmarks establish a foundation for algorithm comparability and represent current community standards in autonomous endovascular navigation research; however, they intentionally focus on fundamental navigation capabilities and do not encompass the full complexity of clinical interventions. As algorithmic capabilities advance, these benchmarks will require systematic extension to address additional challenges. The focus on the anatomy of the aortic arch for neurovascular interventions, while clinical practice requires navigation across diverse vascular territories (coronary, peripheral, neurointerventional). The instrument selection process and intra-interventional instrument exchanges are currently not modeled, while clinical procedures often require switching between different guidewires, catheters, or other instruments mid-procedure based on encountered anatomy or procedural requirements. The repositioning of the imaging system is also not addressed — in clinical practice, physicians frequently adjust C-arm positioning to optimize instrument visibility and reduce anatomical overlap during navigation, which represents a critical skill for complex interventions. Furthermore, benchmarks providing simulated fluoroscopy images as feedback are missing and should be implemented in the future to provide an opportunity to compare control algorithms in an end-to-end manner.

6.3. *stEVE — simulated EndoVascular environment*

The framework *stEVE* offers a modular approach to compose endovascular interventions and embed them in reinforcement learning environments. Its ease of adaptation for individualized intervention and feedback requirements can significantly facilitate the development of robotic assistance systems in endovascular interventions. The digital nature of the simulation framework enables reproducibility of computational results across institutions — identical environment scenarios produce identical simulation outcomes regardless of the research group. This reproducibility is essential for meaningful algorithm comparison in the research community.

The modular design allows for the incorporation of extensions and additional features, potentially introducing new functionalities such as manipulation of the position and orientation of the fluoroscopy system to optimize instrument visibility or force measurement by analyzing the contacts of instrument and vessel wall to provide force feedback. *stEVE* can be adapted to support other endoluminal interventions by implementing the required organ lumen and instruments and adapting the simulation script to the new setup. For instance, with the SOFA simulation engine it is possible to include magnetic continuum robots [51] or cable-driven instrument [52]. Furthermore, complex but interesting future developments include tissue deformation, tissue inhomogeneities, and motions due to heartbeat or breathing. Integrating tissue inhomogeneities can be achieved by adding rigid calcifications to the lumen of the vessel tree. Vessel deformation is highly relevant for interventions such as endovascular aortic repair [53]. Considering deformation during interventions can improve radiation exposure, reduce the need for contrast agents, and shorten the procedure time [54]. This could be implemented by including a deformation model for the vessel mesh, allowing for vessel flexibility, albeit with increased computational demands. Heartbeat effects could be simulated by applying a force field on the instrument that exerts force in the direction of blood flow, while breathing motion could be simulated by applying cyclic movement and deformation patterns to the vessel system. Additionally, while our simulation demonstrates qualitative validity through successful sim-to-real transfer, the lack of comprehensive quantitative validation of simulation fidelity against clinical measurements is an important limitation. Future work should include systematic comparison with clinical force data, high-precision instrument tracking, and standardized phantom experiments to establish quantitative accuracy metrics for the simulation framework.

7. Conclusion

This work introduces the simulation framework *stEVE*, tailored for endovascular interventions with the primary purpose of streamlining and supporting future research efforts in the field of robotic assistance systems in endovascular interventions.

The introduced benchmarks are designed to enhance the comparability of research in the domain of learning-based autonomous endovascular navigation. They serve as a common experimental setup that fosters a baseline for researchers to evaluate and advance their approaches.

The benchmarks establish a common framework for comparing research in learning-based autonomous endovascular navigation. By providing a consistent experimental foundation, they enable researchers to systematically evaluate and iteratively improve their navigation approaches.

The experimental results demonstrate the feasibility of *stEVE* and underscore its potential to transfer simulation-trained controllers to real-world scenarios. However, they also reveal areas that offer opportunities for further enhancement and exploration in future research endeavors, including the concurrent manipulation of two concentric instruments and improvements in adaptation to patient individual vessel geometries.

CRediT authorship contribution statement

Lennart Karstensen: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Harry Robertshaw:** Writing – review & editing, Validation, Investigation, Data curation. **Johannes Hatzl:** Writing – review & editing, Validation. **Benjamin Jackson:** Writing – review & editing, Validation, Data curation. **Jens Langejürgen:** Writing – review & editing, Resources, Project administration.

Katharina Breininger: Writing – review & editing, Supervision. **Christian Uhl:** Writing – review & editing, Validation. **S.M. Hadi Sadati:** Writing – review & editing, Validation, Conceptualization. **Thomas Booth:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Christos Bergeles:** Writing – review & editing, Funding acquisition, Conceptualization. **Franziska Mathis-Ullrich:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Copilot (Microsoft, Redmond, Washington, United States), ChatGPT (OpenAI, San Francisco, United States) and DeepL Write (DeepL SE, Cologne, Germany) in order to improve grammar and readability of this paper. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Benchmark intervention parameters

See Tables A.1–A.3.

Appendix B. Domain gap challenges

See Table B.1.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compbiomed.2025.110844.

Table A.1
BasicWireNav parameters.

Component	Parameter	Value
VesselTree	class	eve.intervention.vesseltree.VMR
	model	"0011_H_AO_H"
	insertion_vessel_name	"aorta"
	insertion_point_idx	−1
	insertion_direction_idx_diff	−2
	approx_branch_radii	5
	rotate_yzx_deg	[0, 133, 0]
Device	class	eve.intervention.device.JShaped
	name	"guidewire"
	velocity_limit	(35, 3.14)
	length	450
	tip_radius	12.1
	tip_angle	$0.4 \times \pi$
	tip_outer_diameter	0.7
	young_modulus_tip	17e3
	mass_density_tip	0.000021
	collis_edges_per_mm_tip	2
	beams_per_mm_tip	1.4
	straight_outer_diameter	0.89
	young_modulus_straight	80e3
	mass_density_straight	0.000021
	collis_edges_per_mm_straight	0.1
	beams_per_mm_straight	0.5
	Poisson_ratio	0.49
	visu_edges_per_mm	0.5
Simulation	class	eve.intervention.simulation.SofaBeamAdapter
	friction	0.1
Fluoroscopy	class	eve.intervention.fluoroscopy.TrackingOnly
	image_frequency	7.5
	image_rot_zx	[0, 0]
Target	class	eve.intervention.target.CenterlineRandom
	threshold	5
	branches	["btrunk", "carotid", "rt_carotid", "subclavian"]

Table A.2
ArchVariety parameters.

Component	Parameter	Value
VesselTree	class	eve.intervention.vesseltree.AorticArchRandom
	episodes_between_change	1
	scale_diameter_array	[0.85]
	arch_types_filter	[eve.intervention.vesseltree.ArchType.I]

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Table A.2 (*continued*).

Component	Parameter	Value
Device	class	eve.intervention.device.JShaped
	name	"guidewire"
	velocity_limit	(35, 3.14)
	length	450
	tip_radius	12.1
	tip_angle	$0.4 \times \pi$
	tip_outer_diameter	0.7
	young_modulus_tip	17e3
	mass_density_tip	0.000021
	collis_edges_per_mm_tip	2
	beams_per_mm_tip	1.4
	straight_outer_diameter	0.89
	young_modulus_straight	80e3
	mass_density_straight	0.000021
	collis_edges_per_mm_straight	0.1
	beams_per_mm_straight	0.5
	Poisson_ratio	0.49
	visu_edges_per_mm	0.5
Simulation	class	eve.intervention.simulation.SofaBeamAdapter
	friction	0.1
Fluoroscopy	class	eve.intervention.fluoroscopy.TrackingOnly
	image_frequency	7.5
	image_rot_zx	[25, 0]
Target	class	eve.intervention.target.CenterlineRandom
	threshold	5
	branches	["lcca", "rcca", "lsa", "rsa", "bct", "co"]

Table A.3
DualDeviceNav parameters.

Component	Parameter	Value
VesselTree	class	eve.intervention.vesseltree.FromMesh
	mesh	".../vessel_architecture_collision.obj"
	insertion_position	[65.0, −5.0, 35.0]
	insertion_direction	[−1.0, 0.0, 1.0]
	rotation_yzx_deg	[90, −90, 0]
	scaling_xyz	[1.0, 1.0, 1.0]
Device1	class	eve.intervention.device.JShaped
	name	"mic_guide"
	velocity_limit	(35, 3.14)
	length	900
	tip_radius	12.1
	tip_angle	$0.4 \times \pi$
	tip_outer_diameter	0.36
	young_modulus_tip	1e3
	mass_density_tip	0.000005
	collis_edges_per_mm_tip	2
	beams_per_mm_tip	1.4
	straight_outer_diameter	0.36
	young_modulus_straight	1e3
	mass_density_straight	0.000005
	collis_edges_per_mm_straight	0.1
	beams_per_mm_straight	0.6
	Poisson_ratio	0.49
	visu_edges_per_mm	0.5
Device2	class	eve.intervention.device.JShaped
	name	"mic_cath"
	velocity_limit	(35, 3.14)
	length	900
	tip_radius	12.1
	tip_angle	$0.4 \times \pi$
	tip_outer_diameter	0.6
	tip_inner_diameter	0.57
	young_modulus_tip	1e3
	mass_density_tip	0.000005
	collis_edges_per_mm_tip	2

(*continued on next page*)

Table A.3 (*continued*).

Component	Parameter	Value
Device2	beams_per_mm_tip	1.4
	straight_outer_diameter	0.7
	straight_inner_diameter	0.57
	young_modulus_straight	1e3
	mass_density_straight	0.000005
	collis_edges_per_mm_straight	0.1
	beams_per_mm_straight	0.6
	Poisson_ratio	0.49
	visu_edges_per_mm	0.5
Simulation	class	eve.intervention.simulation.SofaBeamAdapter
	friction	0.001
Fluoroscopy	class	eve.intervention.fluoroscopy.TrackingOnly
	image_frequency	7.5
	image_rot_zx	[20, 5]
Target	class	eve.intervention.target.CenterlineRandom
	threshold	5
	branches	["LCCA", "LVA", "RCCA", "RVA"]

Table B.1
Potential domain gap challenges and potential solutions.

Challenge	Description	Potential solutions
Physics and Mechanics Domain Gaps		
Instrument-Vessel interaction	Simulation utilizes simplified friction models	Domain Randomization [55] & Learning invariances [56]
Tissue deformation	Simulation employs rigid vessel walls	Perfect Simulator & Domain Randomization [55]
Fluid dynamics	Absence of blood flow effects on instrument behavior	Perfect Simulator [55]
Instrument Characteristics	Real World instrument characteristics might differ from simulation	Domain Randomization [55] & Learning invariances [56]
Pulsatile motion	Missing heartbeat-induced vessel movement and pulsatile blood flow	Data Augmentation [56]
Respiratory motion	Absence of breathing-induced organ displacement and vessel deformation	Data Augmentation [56]
Instrument buckling	Simplified models may not capture complex buckling and kinking behavior	Reward Design [55]
Sensing and Perception Domain Gaps		
Fluoroscopic imaging artifacts	Perfect tracking coordinates vs. noisy fluoroscopic images	Data Augmentation [56]
Visual perception challenges	Sometimes limited instrument visibility in the real world	State representation [55] & Encoding Inductive Biases & Data Augmentation [56]
Temporal delays	Real-world imaging systems have acquisition and processing delays	Data Augmentation [56]
Motion artifacts	Patient movement and breathing cause image blurring and coordinate shifts	Data Augmentation [56]
Calibration drift	Imaging system calibration changes over time, affecting coordinate accuracy	State representation [55] & Encoding Inductive Biases & Data Augmentation [56]
Environmental and Operational Domain Gaps		
Patient variability	Missing pathological conditions and absence of patient-specific tissue properties	Domain Randomization [55] & Learning invariances [56]
Equipment variations	Current Benchmarks employ reduced set of simple curve instruments	Domain Randomization [55] & Learning invariances [56]
Imaging repositioning	Complex interventions require dynamic C-Arm repositioning	Hierarchical RL [55]
Contrast Agent Injection	Absence of contrast injection for determining vessel geometry	Perfect simulator & Hierarchical RL [55]

Data availability

Simulation framework, benchmarks and training scripts are available on GitHub: https://github.com/lkarstensen.

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